LAMMPS official dem03-hertz-tsuji-wall

Sphere-wall rebound and restitution | 29 October 2026

PASS	Four main runs evaluated using the unchanged official selected-time analytic criteria; source tag unvetted.

Official quantities, errors and acceptance limits

Quantity	Reference / limit	CPU dt	GPU dt	CPU dt/2	GPU dt/2
Rebound vx (m/s)	-1 relative limit 0.005	-0.9997437 err 0.0002563 PASS	-0.9997437 err 0.0002563 PASS	-0.9997946 err 0.0002054 PASS	-0.9997946 err 0.0002054 PASS

Check time=7e-05 s; original step 7000. Half dt uses twice the check step at the same time. Each cell shows the native value and its own absolute/relative error; a reference limit is not a same-code epsilon.

Original bodies and source-to-native mapping

Body	Radius / density	Initial xyz (m)	Initial v (m/s)	Initial spin (rad/s)
sphere	R=0.0025 rho=2700	(-0.0026, 0, 0)	(2, 0, 0)	(0, 0, 0)

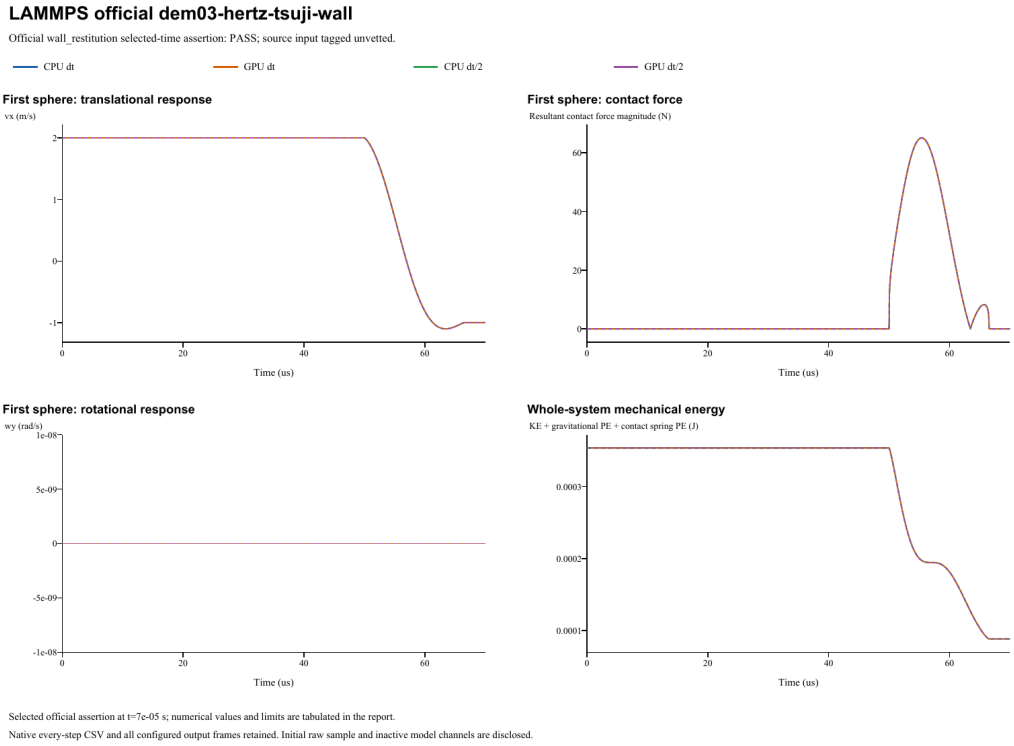
Hertz Fn_elastic=(4/3)*E_eff*sqrt(R_eff)*delta^1.5; native E_eff=3.927729772e+10 Pa. Stored linear kn is inactive. Automatic damping uses the harmonic material restitution and the active normal/tangential stiffness; stored manual dashpots are inactive. Tangential inputs: kt=0 N/m (explicit in linear mode), mu=0.4. Rolling/twist coefficients: kr=0, mu_r=0, ktwist=0, mu_twist=0.

Gravity=(0, 0, 0) m/s^2. Base/half dt=1e-08/5e-09 s; 7000/14000 steps. The pinned wall mixing helper gives E_eff=E/[2*(1-nu^2)]; both native materials retain source E=7e10 Pa. Two previous doubled-E base attempts are preserved as superseded diagnostics and excluded from this four-run verification. Exact Tsuji coefficient mapping: native beta=sqrt(2/5)*P6(e_source); native material restitution is the inverse beta transform, not a fitted rebound. Official analytic target remains e_source. Normal force prefactor and damping at every overlap match.

Pinned LAMMPS source

Official input | Analytic checker | Parameter mapping.
Commit 29dd3fd7642a6fe9e8b4dd99120df4068b221250; input blob b0383cb68ec47ab673287e47b3a0d35768ffbe7d. Official tag: unvetted. LAMMPS was not rerun locally. Physical inputs were not fitted to native results.

LAMMPS native histories and refinement



CPU/GPU and half-step comparison

Quantity	CPU dt error	CPU dt/2 error	Error type / limit
Rebound vx (m/s)	0.0002563	0.00020544	relative: 0.005

CPU/GPU whole-history maximum position / velocity / spin / force differences: 0 m / 4.44e-16 m/s / 0 rad/s / 1.42e-14 N.

Maximum individual impulse / position-drift / velocity-kick residuals: 1.95e-18 kg m/s / 2.17e-19 m / 2.19e-16 m/s. Gravity is added separately to saved native contact forces.

Complete results and verified scope

All 42,004 native frames are retained; main frame counts=[7001, 7001, 14001, 14001]. Every numerical-step particle/contact CSV, four loadable main projects and raw initial samples remain preserved. Output cadence is recorded separately from integration dt. Energy/history diagnostics do not replace the official acceptance quantities.

Run manifest | Analysis | Official metrics | Independent audit | Full native ZIP | Frame hashes

Source checkpoints: Source checkpoint states are supplemental regression diagnostics. Their differences are reported directly; the YAML same-code epsilon is not imposed as an independent cross-software acceptance limit.

A stable helper executable is statically linked to the current native CPU/CUDA libraries. Historical GUI/library byte identity is not asserted. Scope is the active source physics and the selected official criteria; inactive options remain outside this test.